# Phase D4 Cross-Pair Analysis

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## Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

## Abstract

Phase D4 of the CKS derivation note series formalizes twenty-six composition pairs across the inter-Self coordination architecture introduced in Paper 3. Each pair names a specific interaction between two Paper 3 commitments and identifies what that interaction produces or requires. This note, D4.28, performs cross-pair synthesis: it organizes the twenty-six pairs into four functional clusters, identifies three key inter-pair interactions that reveal structural dependencies across the pair set, and identifies the three most foundational pairs on the basis of their architectural leverage. D4.28 establishes that Phase D4's pairs do not constitute an arbitrary list of pairwise observations — they constitute a coherent governance architecture with internal structure, dependency ordering, and identifiable load-bearing commitments. D4.29 (key claims index) and D4.30 (phase closure) follow.

## 1. Why cross-pair analysis is needed

Twenty-six composition pairs, read in sequence from D4.02 through D4.27, can be read as a list. The list form is sufficient for defensive-publication purposes — each pair establishes prior art for a specific composition claim — but it obscures what the pairs collectively say. Cross-pair analysis asks a different question: given all twenty-six pairs together, what is the governance architecture they collectively describe?

The answer requires two analytic moves. The first is clustering: grouping pairs by the governance function each pair performs, so that pairs solving the same class of governance problem can be examined as a group rather than as isolated observations. The second is interaction tracing: identifying pairs that are not logically independent, because one pair's commitment is used as a solved primitive by another pair, or because one pair's output is the validation mechanism for another pair's calibration. Both moves are necessary for a complete reading of Phase D4 as a governance architecture.

This note performs both.

## 2. Four composition pair clusters

The twenty-six pairs divide into four functional clusters. The division is by governance function — what problem each pair solves — not by which Paper 3 claim a pair derives from. Pairs from different claims appear in the same cluster when they solve the same class of governance problem.

### Cluster 1 — Governance Integrity Pairs

Governance Integrity pairs share a common structure: each pair formalizes the commitment that a scenario-specific pressure — an active event, an emergency dissolution, a mid-event withdrawal, a high-frequency operating context — does not justify relaxing a core governance commitment. The pairs in this cluster are not about how governance is performed under normal conditions; they are about what governance commitments remain binding when circumstances create pressure to suspend them.

**D4.05 (Exit Rights + Active Events)** — the right to exit a shared substrate does not suspend the governance rules that apply to dissolution. An active event makes dissolution more complex; it does not make dissolution governance optional.

**D4.14 (Emergency Dissolution + Evolution Feed)** — an emergency dissolution does not suspend the requirement that the evolution feed be governed. Emergency status creates speed pressure; it does not eliminate the governance commitment to capturing what the event produces for downstream learning.

**D4.15 (Partial Withdrawal + Conflict Registry)** — a partial withdrawal from the shared substrate does not permit deletion of conflict records. Withdrawal governance operates at the participation-state layer; conflict records belong to the substrate's audit architecture and are not within the scope of what a withdrawal removes.

**D4.24 (Instinct/Reasoning + Exchange Bounding)** — the theoretical framing from the instinct/reasoning separation supplies the ground for what exchange bounding protects. Bounding is not an arbitrary restriction on what aspects can exchange; it is the governance expression of the architectural commitment that certain commitments are non-negotiable at the exchange boundary.

**D4.25 (Determinism + Non-Delegation)** — automated governance — specifically, permitting a non-human process to make governance decisions without pre-authorization — breaks the determinism contract. Pre-authorized labor maintains determinism because the authority for the labor's scope was established by human governance before execution. The pair establishes that determinism is not a property of execution mechanics alone; it is a property of the governance architecture within which execution occurs.

These five pairs are the governance integrity backbone of Phase D4. Any implementation of the inter-Self coordination architecture that treats scenario-specific pressure as a justification for governance relaxation violates the commitments these pairs formalize.

### Cluster 2 — Governance Quality Pairs

Governance Quality pairs govern how governance is performed well, rather than whether governance applies. They address calibration, documentation standards, review discipline, and the feedback loops that determine whether governance monitoring is accurately configured.

**D4.06 (High-Frequency + Health Monitoring)** — health monitoring thresholds must be frequency-adjusted. What constitutes a governance health signal in a low-frequency relationship differs from what constitutes the same signal in a high-frequency relationship; fixed thresholds applied across both contexts produce miscalibrated governance.

**D4.10 (Regulatory Audit + Documentation Standards)** — documentation standards for regulatory audit are not identical to documentation standards for internal governance review. The pair formalizes the composition requirement: both documentation regimes must be satisfied, and satisfying one does not substitute for the other.

**D4.22 (Versioning + Process Disputes)** — versioning is the substrate-level mechanism that makes process disputes resolvable. When a dispute arises about which governance process applied at a specific moment, the version history answers the question. The pair formalizes that versioning is not merely a record-keeping convenience; it is the dispute-resolution mechanism for a specific class of governance conflicts.

**D4.26 (Health Indicators + Post-Mortem)** — post-mortem review of completed FAI events is the calibration mechanism for health indicators. The pair formalizes that health monitoring (D4.06) requires a feedback loop: the post-mortem produces the evidence base for determining whether current health thresholds are correctly configured.

### Cluster 3 — Relationship Architecture Pairs

Relationship Architecture pairs govern the structural commitments that sustain multi-party relationships over time. They address how standing configurations work, how parent-child governance document hierarchies operate, how asymmetric relationships are governed, and how high-frequency relationships are architecturally differentiated from episodic ones.

**D4.03 (Standing Configurations + Multi-Self Events)** — standing configurations govern how multi-Self events inherit default parameters. The pair formalizes that multi-Self events do not begin from a blank governance slate; they begin from the standing configuration each participating Self brings, with the event's governance configuration layered above those standing defaults.

**D4.13 (Cross-Organizational Agreement + Standing Configuration)** — a cross-organizational agreement establishes the governance parent from which participating Selves' standing configurations derive their authority. The pair formalizes the parent-child relationship: standing configurations are children of the cross-organizational agreement, and the agreement's terms govern when and how standing configurations can be modified.

**D4.16 (Asymmetric Selves + Standing Configurations)** — when participating Selves have asymmetric capability profiles or governance structures, standing configurations must specify how that asymmetry is handled at the governance layer. The pair formalizes that asymmetry is a first-class architectural condition, not an edge case that standing configurations can ignore.

**D4.17 (High-Frequency + Cross-Organizational Agreement)** — high-frequency inter-Self relationships require that the cross-organizational agreement explicitly address frequency-specific governance obligations. The pair formalizes that frequency is a governance dimension, not merely an operational one; agreements that do not address frequency leave a governance gap that becomes visible as those relationships scale.

**D4.23 (Multi-Domain Self + Cross-Organizational Agreement)** — when a participating Self operates across multiple content domains, the cross-organizational agreement must address how multi-domain scope interacts with the shared substrate's configuration. The pair formalizes that domain coverage is a governance-relevant property of participating Selves, not merely an operational description.

### Cluster 4 — Learning and Evolution Pairs

Learning and Evolution pairs govern how FAI events produce durable governance improvement. They address DNA absorption, aspect restructuring, conflict carry-through, post-mortem records, and the mechanisms by which what governance learns from one event is available to future events.

**D4.08 (DNA Absorption + Aspect Restructuring)** — when a Self absorbs DNA-layer content from a shared substrate event, that absorption may require restructuring the receiving Self's aspects. The pair formalizes that absorption is not a passive copy operation; it is a governance event that may trigger architectural adjustment.

**D4.09 (Conflict Carry-Through + Vertical Evolution)** — conflicts that arise within a shared substrate event and are not resolved before dissolution must carry through to the receiving Self's home substrate, where they enter the vertical evolution process. The pair formalizes that unresolved conflicts are not discarded at dissolution; they are governance artifacts with a defined downstream destination.

**D4.11 (Knowledge Transfer + DNA Absorption)** — knowledge transfer across the inter-Self boundary and DNA absorption are distinct operations with distinct governance requirements. The pair formalizes that a Self can receive transferred knowledge without absorbing the DNA-layer structure that produced it, and vice versa; the two operations should not be conflated in governance configuration.

**D4.12 (Competitive Intelligence + Post-Mortem)** — post-mortem review of FAI events may produce competitive intelligence for participating organizations. The pair formalizes that this is a governance-relevant outcome of post-mortem processes, not an incidental side effect; governance configuration must address how competitive intelligence is handled within the post-mortem record.

**D4.19 (Trust Calibration + DNA Absorption)** — the trust level governing what a Self absorbs from a shared substrate event is itself a governance-configurable parameter. The pair formalizes that trust calibration operates at the absorption boundary, and that absorption depth is a function of governance-configured trust, not of technical transfer capacity.

**D4.20 (Observational Studies + Records)** — observational study of inter-Self coordination patterns requires that records be structured to support retrospective analysis. The pair formalizes that record quality is a governance commitment, not merely a documentation preference; records that cannot support observational study fail a governance standard.

## 3. Three key pair interactions

The clusters establish what each group of pairs does. The key pair interactions establish how the pairs depend on each other — where one pair's commitment is the mechanism another pair uses, or where one pair's output is another pair's input. Three interactions are architecturally significant.

### D4.02 enables many pairs

D4.02 (Floor Governance + Time-Sensitive Operations) establishes pre-authorization as the canonical resolution of the speed-governance tension. The problem the pair addresses — how governance commitments remain binding when time-sensitive conditions create pressure to bypass them — arises in multiple other pairs. D4.04 (Competition + Preserve), D4.14 (Emergency + Evolution Feed), D4.17 (High-Frequency + Agreement), and D4.27 (Non-Specialist + Floor) each encounter a variant of the same problem. Each resolves it by invoking pre-authorization. D4.02 is the pair that establishes pre-authorization as the available resolution — without D4.02, each downstream pair would need to independently construct the same solution.

This makes D4.02 the architectural enabler for the pre-authorization cluster within Phase D4. Any subsequent governance design that addresses speed-governance tension without the pre-authorization pattern will need to provide an alternative — and will need to address why the alternative is preferable to the pattern D4.02 establishes.

### D4.13 is the parent pair for D4.16 and D4.17

D4.13 (Cross-Organizational Agreement + Standing Configuration) establishes the parent-child governance document hierarchy: the cross-organizational agreement is the parent governance document from which standing configurations derive their authority. D4.16 (Asymmetric Selves + Standing Configurations) and D4.17 (High-Frequency + Cross-Organizational Agreement) both operate within this hierarchy. D4.16 asks how the parent-child hierarchy handles asymmetric participating Selves; D4.17 asks how the parent-child hierarchy addresses high-frequency operating conditions. Neither pair can be fully understood without D4.13's hierarchy as its structural premise.

The dependency is not merely conceptual. A governance designer implementing D4.16's asymmetric-Self provisions or D4.17's high-frequency provisions without first implementing D4.13's parent-child hierarchy will encounter a structural gap: the asymmetric and high-frequency commitments presuppose a governance parent from which they inherit authority. D4.13 is the pair that creates that parent.

### D4.26 validates D4.06

D4.06 (High-Frequency + Health Monitoring) establishes that health monitoring thresholds must be frequency-adjusted. The commitment is sound in principle but cannot be acted on without a mechanism for determining whether current thresholds are correctly calibrated. D4.26 (Health Indicators + Post-Mortem) supplies that mechanism: the post-mortem review of completed events provides the evidence base against which current health indicator thresholds can be assessed and revised.

The interaction makes explicit a feedback loop that would otherwise be implicit. D4.06 configures health monitoring for frequency; D4.26 tells governance how to know whether D4.06's configuration is correctly set. Without D4.26, D4.06's frequency-adjusted thresholds are set once and held; there is no governed mechanism for discovering that a threshold is miscalibrated. With D4.26, miscalibration is detectable through post-mortem evidence and correctable through governed threshold revision.

## 4. Three most foundational pairs

Cross-pair analysis supports an assessment of which pairs are most foundational — not most complex, nor most novel, but most load-bearing as prior art claims. The three most foundational pairs are the pairs that establish principles which any serious governance architecture for rapid, sustained, or automated inter-Self coordination must address.

### D4.02 — Floor Governance and Time-Sensitive Operations

D4.02 is foundational because it names and resolves the most universal problem in governed coordination under time pressure: how does a governance commitment remain binding when speed creates pressure to bypass it? The answer — pre-authorization, where human governance establishes in advance the conditions under which a time-sensitive action is authorized, so that the action at execution time is governed, not ungoverned — is the architectural move that makes governed rapid coordination possible without degrading governance to a post-hoc approval process.

Any system that coordinates rapidly under governance must either adopt the pre-authorization pattern or provide an alternative. D4.02 is the prior-art claim that the pre-authorization pattern is available, has been named, and has been applied to this class of problem. A designer arguing for an alternative must argue against D4.02 explicitly.

### D4.13 — Cross-Organizational Agreement and Standing Configuration

D4.13 is foundational because it names and formalizes the parent-child governance document hierarchy for sustained inter-party relationships. In any architecture that sustains multiple parties under governance over time, the question arises: when a specific relationship's governance configuration differs from a general framework, which governs? D4.13's answer — the cross-organizational agreement is the parent; standing configurations are children that derive authority from the parent and are bounded by it — is the architectural move that gives the relationship governance architecture coherence across multiple participating pairs, across time, and across changes in specific relationship configurations.

Any sustained multi-party governance architecture must address the parent-child hierarchy question. D4.13 is the prior-art claim that this hierarchy is available, has been named, and has been applied to the inter-Self relationship governance context.

### D4.25 — Determinism and Non-Delegation

D4.25 is foundational because it names and formalizes the link between human governance authority and governance auditability at the automation boundary. The problem the pair addresses — when an automated process makes governance decisions, the audit trace becomes ambiguous: did human governance authorize this specific decision, or only the general delegation that produced the automated process? — is the central problem of governed AI automation. D4.25's answer — automated governance breaks determinism; pre-authorized labor maintains it; the distinction between the two is the governance architecture's accountability boundary — connects two of the deepest commitments in the CKS architecture: that governance is traceable to human authority, and that the substrate records what was authorized, not merely what was executed.

Any governed AI automation architecture must address this question. D4.25 is the prior-art claim that the pre-authorized-labor solution is available, has been named, and has been applied to the governance-determinism boundary in inter-Self coordination.

## 5. Phase D4 as a governance architecture

The four clusters, three key interactions, and three foundational pairs together establish what Phase D4's twenty-six pairs collectively say. They are not an arbitrary enumeration of pairwise composition observations. They describe a governance architecture with identifiable structure:

A **governance integrity backbone** (Cluster 1) that holds core commitments binding regardless of scenario-specific pressure.

A **quality improvement layer** (Cluster 2) that calibrates how governance is performed and closes the feedback loops needed for calibration to improve over time.

A **relationship structure** (Cluster 3) that governs how sustained inter-party relationships are architecturally configured, hierarchically organized, and differentiated by operating conditions.

A **learning mechanism** (Cluster 4) that governs how FAI events produce durable improvements in governance, not just in operational outcomes.

The interactions (Section 3) reveal that this architecture has internal dependency structure: D4.02 enables the pre-authorization layer on which multiple pairs depend; D4.13 creates the hierarchy within which D4.16 and D4.17 operate; D4.26 closes the calibration loop for D4.06. The foundational pairs (Section 4) identify where the architecture's prior-art protection is strongest: at the principles no governed coordination designer can avoid.

D4.29 (key claims index) and D4.30 (phase closure) follow.